

Fracking Health Impacts on Humans

Snigdha Gurram
Department of Environmental
Studies

San Jose State University

Introduction:

The importance of this research is to understand whether fracking has serious health impacts for children under the age of 18 in California and Texas as well as the relationship between Hydraulic Fracking sites and Cancer Rates in California

RQ1: Is there a difference in the number of kids at risk of human health impacts due to fracking between California and Texas?

fracking between California and Texas?

Alternative Hypothesis: There is a different in the number of kids at risk of human health impacts due to fracking between California and Texas.

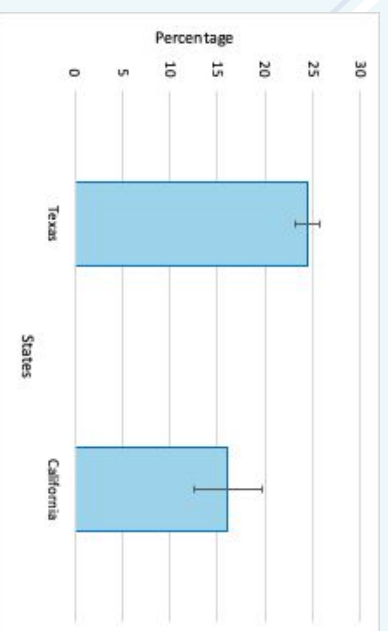


Figure 1. Number of Children at Risk Under 18

RQ2: Is there a relationship between the number of sites and the cancer rates in California?

and the cancer rates in California?

Alternative Hypothesis: There is a relationship between the number of sites and the cancer rates in California.

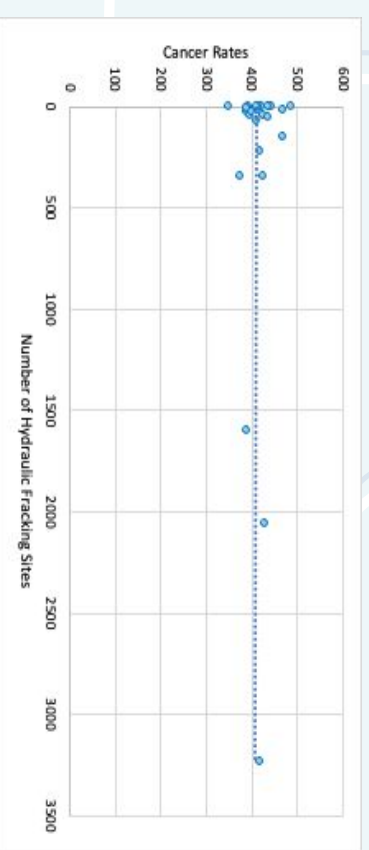


Figure 2. Relationship Between Hydraulic Fracking Sites And Cancer Rates

Methods:

RQ1:

- Randomly select 15 counties for each state
- Number of children at risk of health impacts data from Oil and Gas Threat Map
- T-test or Mann Whitney

RQ2:

- Randomly select 30 counties for California
- Data for the number of cancer cases from californiahealthmaps.org
- Linear Regression or Spearman Rank

Results:

RQ1:

- T-test was used
- p-value of less than 0.05 tells us that this is statistically significant difference
→ P-value = 0.02 < 0.05

RQ2:

- Linear regression was used
- Normally distributed
- Not statistically significant
→ P-value = 0.87
→ Kurtosis: 0.85 & Skew: 0.36

Recommendations:

- Texas should understand the negative health impacts that are happening because of fracking
- Reduce the number of fracking sites in one area
- Make a designated area for only fracking (and other health impacting industries) and have no residential living near it

Conclusion:

With all of this information, it can be seen that there are serious impacts occurring because of hydraulic fracturing that affects more than the environment but also the people in the surrounding area.

Citations:

